

1139 - cloudflare: Cloudflare Reverse Proxies are Dumping Uninitialized Memory - project-zero

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Content

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1139 - cloudflare: Cloudflare Reverse Proxies are Dumping Uninitialized Memory - project-zero - Monorail

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| | | |

| Status: | Fixed | | | Owner: |

taviso@google.com

| | | Closed: | Today | | | Cc: |

taviso@google.com

project-...@google.com

- | |

|

Deadline-90

Finder-taviso

Severity-Critical

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Reported-2017-Feb-17

Vendor-cloudflare

Product-cloudflare

||

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(It took every ounce of strength not to call **this** issue "cloudbleed")

Corpus distillation is a procedure we **use** to optimize the fuzzing we **do** by analyzing publicly available datasets. We've spoken a bit about **this** publicly in the past, **for** example:

```
[https://security.googleblog.com/2011/08/fuzzing-at-scale.html]
(https://security.googleblog.com/2011/08/fuzzing-at-scale.html)
[http://taviso.decsystem.org/making_software_dumber.pdf#page=11]
(http://taviso.decsystem.org/making_software_dumber.pdf#page=11)
```

On February 17th 2017, I was working on a corpus distillation project, when I encountered some data that didn't match what I had been expecting. It's not unusual to find garbage, corrupt data, mislabeled data or just crazy non-conforming data...but the format of the data **this** time was confusing enough that I spent some time trying to debug what had gone wrong, wondering **if** it was a bug in my code. In fact, the data was bizarre enough that some colleagues around the Project Zero office even got intrigued.

It became clear after a **while** we were looking at chunks of uninitialized memory interspersed with valid data. The program that **this** uninitialized data was coming **from** just happened to have the data I wanted in memory at the time. That solved the mystery, but some of the nearby memory had strings and objects that really seemed like they could be **from** a reverse proxy operated by cloudflare - a major cdn service.

A **while** later, we figured out how to reproduce the problem. It looked like that **if** an html page hosted behind cloudflare had a specific combination of unbalanced tags, the proxy would intersperse pages of uninitialized memory into the output (kinda like heartbleed, but cloudflare specific and worse **for** reasons I'll explain later). My working theory was that **this** was related to their "ScrapeShield" feature which parses and obfuscates html - but because reverse proxies are shared between customers, it would affect **all** Cloudflare customers.

We fetched a few live samples, and we observed encryption keys, cookies, passwords, chunks of POST data and even HTTPS requests **for** other major cloudflare-hosted sites **from** other users. Once we understood what we were seeing and the implications, we immediately stopped and contacted cloudflare security.

This situation was unusual, PII was actively being downloaded by crawlers and users during normal usage, they just didn't understand what they were seeing. Seconds mattered here, emails to support on a friday evening were not going to cut it. I don't have any cloudflare contacts, so reached out **for** an urgent contact on twitter, and quickly reached the right people.

[<https://twitter.com/taviso/status/832744397800214528>]
(<https://twitter.com/taviso/status/832744397800214528>)

After I explained the situation, cloudflare quickly reproduced the problem, told me they had convened an incident and had an initial mitigation in place within an hour.

"You definitely got the right people. We have killed the affected services"

****This** bug is subject to a 90 day disclosure deadline. After 90 days elapse**
or a patch has been made broadly available, the bug report will become
visible to the public.

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